

Project Workflow Phases

Phase 1

1. Vendor - Project Initiation & Document Submission

Mark work as completed

Upload all required documents (COC, Photos, Permits, Receipts, Safety Forms, Drawings)

View SLA countdown (7-day deadline)

Receive confirmation numbers

Color-coded alerts for overdue projects

2. Supervisor - Project Creation & Assignment

Create new projects

Assign to vendors

Set SLA deadlines

Approve submitted documents

View project statistics dashboard

Priority-based project management

3. Clerk - Document Validation

Validate document submissions

Check file formats and quality

Flag quality issues (file size, type validation)

Generate confirmation numbers

Send notifications to vendors and supervisors

Approve/reject individual documents

4. Admin - User Access Management

Create/manage user accounts

Assign roles and permissions

Activate/deactivate users

View user directory

Monitor system security

Track user sessions

Phase 2

Vendor - Penalty Notifications & Disputes

View penalty details with amount breakdowns

Submit disputes with detailed explanations

Track penalty status (Draft, Issued, Paid, Waived, Disputed)

See summary cards and penalty history

Clerk - Penalty Documentation Support

View delayed projects requiring documentation

Auto-calculate penalties (delay days × 0.1% per day)

Prepare penalty memos with checklist

Forward documentation to supervisor

Supervisor - Penalty Review & Approval

Color-coded dashboard (GREEN / YELLOW / RED status)

Review pending penalties

Approve/reject with notes

Monitor real-time project status

Team Leader - Secondary Approval & Consistency

Review high-value penalties (≥ \$50,000)

Secondary approval workflow

Department-wide consistency monitoring

Supervisor performance comparison

Analytics dashboard

Sector Manager - Sector-Wide Compliance
Sector-wide compliance dashboard
Monitor 150+ projects and \$25M pipeline
Vendor compliance performance tracking
Strategic reports and trends
Multi-department oversight

Phase 3

1. Supervisor QI Assignment Dashboard
AI-powered QI recommendations (workload + GPS + specialization)
QI team capacity overview with real-time percentages
Pending projects awaiting assignment
Accept or override AI suggestions
Track recent assignments

2. Team Leader Workload Balancing
Multi-supervisor aggregate view
Department-wide QI capacity monitoring
Auto-balance algorithm ($\pm 8\%$ variance target)
Workload distribution by supervisor
Variance tracking and optimization

3. QI Mobile Dashboard
Inspection queue sorted by urgency (URGENT/DUE/SCHEDULED)
Accept or request reassignment
View assignment details
Capacity tracker display
Push notifications integration

4. Clerk QI Coordination
Process reassignment requests
QI calendar management (7-day view)
Auto-calendar integration
Email + push notifications
Schedule conflict resolution

Phase 4

1. QI Digital Inspection Interface
GPS auto-timestamp on arrival
Project-specific digital checklists (Electrical, Civil, Mechanical)
PASS/FAIL marking with photo capture
Voice-to-text notes
Offline mode with sync capability
Digital signature
Auto-generates defect reports when items fail

2. Vendor Defect Corrections Interface
View all defect reports with failed items
Upload corrective action photos
Submit correction notes
Track deadline countdown
Request deadline extensions
Handle multiple failure attempts
Escalation warnings after 3 failures

3. Supervisor Inspection Monitoring
Real-time inspection statistics dashboard
View all inspection records with filters
Approve/reject vendor corrections
Escalate cases after 3 failures to Team Leader
Monitor re-inspection schedule

View detailed inspection reports with photos

4. Clerk Re-Inspection Scheduling

- Schedule re-inspections with same QI
- Coordinate QI calendar
- Auto-notify QI and vendor
- Reschedule inspections
- Archive inspection documents (10-year retention)
- Track focus items (previously failed items)

5. Engineering Aide Coordination Dashboard

- Coordinate re-inspection logistics
- Technical review of complex failures
- Escalate to higher authority
- Monitor active escalations
- Track re-inspection schedule
- Technical evaluation and recommendations

Phase 5

1. Clerk - Pre-Billing Validation

- 6-point validation checklist (inspection approved, documents complete, no pending corrections, penalties calculated, account good standing, no legal holds)
- Document verification (COC, QI Report, receipts, penalty memos)
- Submit for billing workflow
- Real-time validation status

2. Supervisor - Billing Calculation & Approval

- Billing calculation breakdown (contract amount - penalties - deductions = net amount)
- Penalty details display
- Invoice number generation
- Approve & generate invoice workflow
- Auto-updates project status to "Awaiting Payment"

3. Team Leader - Financial Oversight

- High-value invoice review ($\geq$ \$100,000 threshold)
- Financial metrics dashboard (pending review, total value, high-value count)
- Value categorization (Very High $\geq$ \$1M, High $\geq$ \$500K, Medium $\geq$ \$100K)
- Route to Finance with approval

4. Vendor - Invoice & Payment Tracking

- Personal invoice dashboard
- Payment status tracking (Unpaid, Partially Paid, Paid, Overdue)
- Invoice details view
- PDF download capability
- Performance stats (total invoices, pending payment, YTD collected)

5. Sector Manager - Billing Metrics

- Sector-wide billing analytics
- Key metrics (total billed, collected, pending, avg payment cycle)
- On-time billing rate tracking (target: 95%)
- Collection efficiency metrics
- Monthly billing trends visualization
- Report export functionality

Phase 6

1. Clerk - Pre-Billing Validation

- Validates 6 critical checks before billing
- Document completeness verification
- Inspection approval confirmation
- Penalty calculation validation
- Account status checks

Submits validated projects for billing approval

2. Supervisor - Billing Approval

Reviews billing calculations
Displays contract amount minus penalties
Shows detailed penalty breakdown
Approves and generates invoices
Routes to Finance with full documentation

3. Vendor - Payment & Invoice Tracking

Monitors all invoices and payment status
Downloads invoices and payment receipts
Views performance scorecard (on-time rate, quality, SLA compliance)
Tracks payment cycles (Net 30 days)
Color-coded status indicators

4. Team Leader - Billing Oversight

Department-wide billing dashboard
Reviews high-value invoices (>\$100,000)
Monitors total pending billing value
Tracks average payment cycle times
Routes approved invoices to Finance

5. Sector Manager - Billing Metrics & Portfolio

Sector-wide \$25M+ pipeline overview
Vendor performance rankings with renewal recommendations
6-month billing trends visualization
Strategic partnership approvals
Budget vs. actuals tracking

6. Admin - Document Archiving & Audit

10-year retention compliance management
Cloud + local backup tracking
Complete audit trail maintenance
Read-only permissions enforcement
Storage capacity monitoring